

Proactive Master Operation :

A **Proactive Master** is not a specific UVM component or class defined by the IEEE UVM standard. Instead, it is a **behavioral pattern** that describes how a UVM **driver** operates when interacting with the Design Under Test (DUT).

In a UVM verification environment, a driver's behavior depends on the protocol being verified and the role played by the DUT. When a driver operates as a *proactive master*, it **actively initiates protocol transactions** rather than waiting for any request or trigger from the DUT.

Understanding the “Proactive” Nature

The term *proactive* refers to how the driver controls communication on the interface. A proactive driver **does not wait for the DUT to signal or request an operation**. Instead, it takes the initiative by retrieving sequence items from the sequencer and driving them onto the DUT interface according to the protocol timing and rules.

The stimulus flow is entirely controlled by the verification environment:

- Sequences determine **what** transactions are sent
- The driver determines **when and how** the protocol signals are driven

When operating in this mode, the DUT acts as a **passive responder**, reacting to transactions initiated by the driver, assuming the DUT implements the slave side of the protocol.

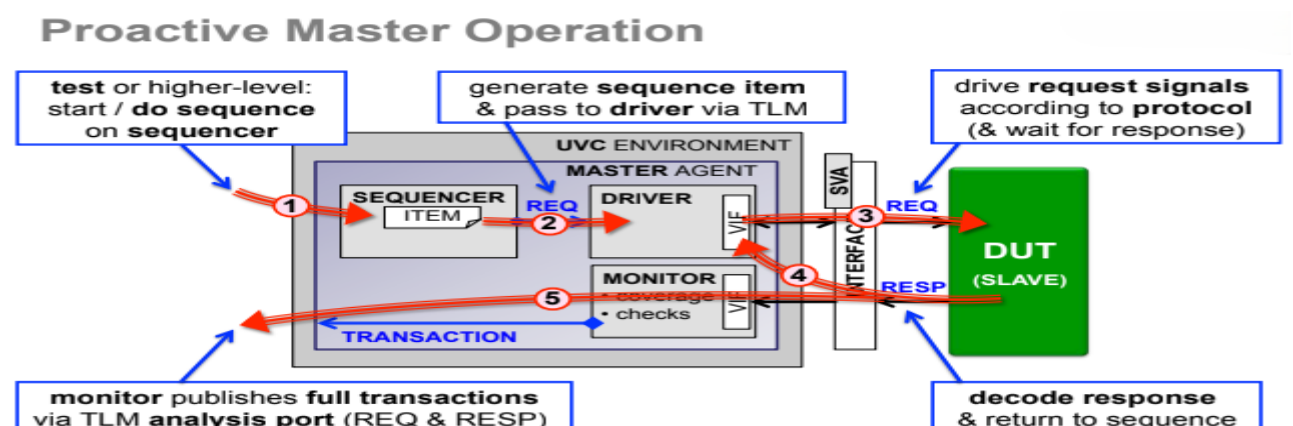
The “Master” Role in Protocol Communication

Most hardware communication protocols are based on a **master – slave relationship**. The master controls the bus, initiates all transactions, and defines the operation type and timing. The slave waits for the master's request and responds accordingly.

When a UVM driver functions as a proactive master, it models the **master side of the protocol**. Examples include:

- A CPU initiating read and write transactions to memory
- An AXI master driving address and control channels
- An I²C master starting transfers to slave devices

In all these cases, the driver **actively drives the interface signals** to initiate communication, while the DUT responds based on protocol requirements.



REACTIVE SLAVE Operation :

A reactive slave in UVM is an agent behavior in which the DUT controls the entire transaction flow by acting as the master, and the verification environment responds only after observing valid activity on the interface. In this model, the slave does not initiate any stimulus on its own. The monitor continuously observes signal-level activity, decodes DUT-initiated requests, and converts them into transactions. These request transactions are forwarded to a sequencer operating in reactive mode, which typically runs a passive or forever sequence that waits for requests rather than generating traffic. The driver then produces and drives the corresponding responses back to the DUT, strictly following protocol timing, ordering, and handshake rules. Both request and response transactions are published through analysis ports for checking, coverage, and scoreboarding. This approach accurately mirrors real hardware slave behavior, keeps responses synchronized with DUT actions, and prevents illegal or premature signaling, making it essential whenever all transactions must be initiated by the master.

Reactive Slave Operation

